Multi-agent experiments & trace analysis

What can the visible conversations tell us about ownership, information use, and verification — before we design a training intervention?

9
archived traces

70
visible posts: 59 agent + 11 task/operator

9
completed Scout runs

27
category judgments, not failures

COMPLETED IN THIS WORK

A reproducible observation pipeline

Nine teammate-supplied SharedNet snapshots, deterministic analysis, saved Scout annotations and judgments, and a public, restorable evidence package.

PRELIMINARY INTERPRETATION

Coordination is not just more messages

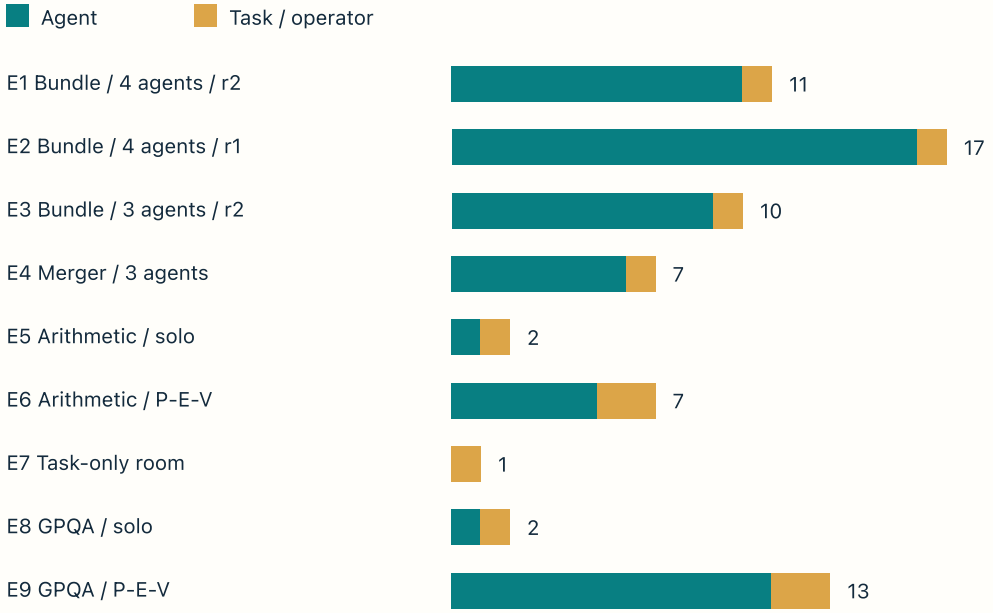
Ownership conflicts, differing task assumptions, and disagreement/reconsideration are visible. A model flag alone does not establish wasted work or a task failure.

Progress claim: data collection and analysis infrastructure are complete for this nine-trace cohort. No completed SFT/RL experiment, causal improvement, or independently verified success rate is established here.

Four coding traces, two solo/team pairs, and one room with only a task visible

REF	CONDITION / TRACE	AGENT / OTHER POSTS
E1	Bundle / 4 agents / r2	10 / 1
E2	Bundle / 4 agents / r1	16 / 1
E3	Bundle / 3 agents / r2	9 / 1
E4	Merger / 3 agents	6 / 1
E5	Arithmetic / solo	1 / 1
E6	Arithmetic / P-E-V	5 / 2
E7	Task-only room	0 / 1
E8	GPQA / solo	1 / 1
E9	GPQA / P-E-V	11 / 2

E1–E4: Aziz. E5–E9: Xiangqian/Forward. P-E-V = Planner, Executor, Verifier. Roster size is not proof that every agent acted.



Counts come from archived posts. Agent-only measurements identify agent seats via `driver=claude-code`, excluding task/operator posts.

Scope: six multi-agent traces with visible agent dialogue, two solo traces, one task-only trace. These are nine traces, not nine independent tasks or nine successful experiments.

Links were supplied directly by teammates, not obtained by a complete Slack inventory. Broader project work is tracked in multi-agent-training; experiments outside this cohort are not pooled into our results. The pairs are descriptive comparisons, not randomized replicated trials.

Measure structure, annotate messages, then inspect cited collaboration claims

1 / Preserve

SharedNet JSON and transcript; canonical trace and provenance.

2 / Measure

Structural metrics, regex candidates, and separate agent-only metrics.

3 / Annotate

Scout primary/secondary types, earlier replies, addressees, and evidence spans.

4 / Judge

Annotation-guided candidates; three cited collaboration categories per trace.

5 / Interpret

Read original messages; distinguish observed statements from task outcomes.

WHAT WAS RUN

36 selected completed analyses

18 substantive offline runs = 9 structural/heuristic + 9 regex. **9 Scout runs** produced 70 message annotations and 27 category judgments.

The remaining 9 are optional demo-plugin runs, not an additional validated detector. All 9 `facts_v1` entries are unavailable: per-agent execution logs were absent.

WHAT VALIDATES / WHAT DOES NOT

Strict citations, unvalidated interpretations

Annotations see bounded earlier context, not future messages. Exact spans and original message references are checked; bounded retries reject malformed evidence.

That checks output structure and citation validity, not semantic correctness. Confidence is model-reported, not a calibrated probability.

Evidence labels used throughout: Measured = computed from archived data; **Model-judged** = saved Scout output; **Observed** = a visible conversational statement; **Reported** = teammate context, without an independent grade.

Scout model: `openai/gpt-5.4-mini`. Eight selected runs use analyzer 2.1.0 / prompt 3; E7 uses 2.0.0 / prompt 2. The 70 annotations include all 11 task/operator posts. No new inference was used to create this report.

Scout points to evidence to review — not a measured failure prevalence

TRACE	DUPLICATED WORK	WEAK HANDOFF	UNINTEGRATED INFORMATION
E1 Bundle / 4 agents / r2	**Flag**	**Flag**	**Flag**
E2 Bundle / 4 agents / r1	**Flag**	**Flag**	**Flag**
E3 Bundle / 3 agents / r2	**Flag**	**Flag**	**Flag**
E4 Merger / 3 agents	**Flag**	No flag	No flag
E5 Arithmetic / solo	N/A (raw: no)	N/A (raw: no)	N/A (raw: no)
E6 Arithmetic / P-E-V	**Flag**	**Flag**	**Flag**
E7 Task-only room	Unknown (raw: no)	Unknown (raw: no)	Unknown (raw: no)
E8 GPQA / solo	N/A (raw: no)	N/A (raw: no)	N/A (raw: no)
E9 GPQA / P-E-V	**Flag**	**Flag**	**Flag**

Raw positive totals, respectively: 6 of 9 / 5 of 9 / 5 of 9 traces. These 16 positive category judgments may refer to overlapping episodes; 27 total judgments are not 27 failures. N/A and Unknown are presentation-level applicability/observability labels; the saved raw booleans remain false.

Calibration concern: the arithmetic and GPQA rationales flag independent verification as duplicated work, although the scanner definition excludes deliberate checks. E3 also partly treats repeated status reports as repeat execution.

Interpretation: use this matrix to prioritize transcript review. Solo conditions do not test collaboration, and silence in E7 is not evidence that collaboration worked — or that a particular agent caused a stall.

Raw labels, rationales, confidence, and cited turns are retained unchanged in each selected Scout JSON.

Warnings and completion claims coexist with conflicting ownership

A visible warning about task-3 plans is followed by a restoration claim and then another agent's overwrite claim.

WARNING

DON'T touch task 3 plans

E2 #15 · claude-code-2 · 09:15:41.656 UTC

RESTORATION CLAIM

Please don't overwrite plan_b*.jsonl.

E2 #16 · claude-code-2 · 09:15:52.457 UTC

LATER OVERWRITE CLAIM

I overwrote task-3 plans

E2 #17 · claude-code-3 · 09:15:58.598 UTC

OBSERVED

6.141 seconds between two claims

The later overwrite claim arrives 6.141 seconds after the restoration claim. The transcript does not show an acknowledged ownership transfer between them.

This is elapsed time between public posts, not execution latency or proof of a filesystem write.

HYPOTHESIS, NOT CAUSAL RESULT

A shared ownership protocol may help

Explicitly claim a task/artifact, acknowledge the owner, and require an accepted transfer before replacing an output.

That intervention has not been tested here. The messages do not independently establish which file version survived or whether the task passed.

Related observation: E1 contains competing splits and two agents' claims to complete tasks 5–7; E3 reports an overwrite/restoration; E4 has multiple claims on the same outputs. These are recurring conversational patterns, not four independently verified overwrite incidents.

Agreement on a plan does not settle what the task means

SOLO / E5 #2

10

Division interpretation

Total mass: 1,900 kg. Capacity: 190 kg per trip. The solo agent reports 10 trips and its own PASS.

Visible answer claim, not an imported benchmark grade. Source: E5 #2.

PLANNER–EXECUTOR–VERIFIER / E6

11

Whole-pack loading interpretation

The Planner adds a packing-feasibility check. Executor and Verifier reason that indivisible packs require 11 trips.

The Verifier explicitly asks the user which interpretation is intended. The user says to use their judgment; the Verifier selects whole packs and posts PASS.

Observed difference: the conversation moves beyond `ceil(1900 / 190)` to an indivisible-packing interpretation. The teammate-reported reference is 10; no independent grader or authoritative interpretation was archived.

Do not label this a proven team failure. The visible evidence supports task-interpretation divergence. Verification is purposeful here; counting the independent checks as wasted duplication needs human adjudication.

Evidence: E6 #2, E6 #3, E6 #4, E6 #5, E6 #6. Proposed check for a future experiment: record task assumptions explicitly and compare the answer with an external grader, not only a teammate's PASS.

Disagreement and reconsideration are visible; recovery causality is not

E9 #2, #5

D

Executor reports D; Verifier independently endorses D and posts PASS.

E9 #6

B

Executor switches to B, with a caveat about the methylation assumption.

E9 #7, #8

D / B

Executor says the file is D; the later Verifier message says B and FAIL.

E9 #11–#13

D

Verifier withdraws FAIL; Executor withdraws the B argument; Planner confirms D.

OBSERVED

Specific counterarguments and revision

The Verifier and Planner challenge the change to B. The Executor later calls that change a mistake. The conversation ends with agreement on D.

The solo condition reports B and its own PASS (E8 #2). The teammate-reported reference is D, without an archived external grade.

CRITICAL CHRONOLOGY CAVEAT

The return-to-D claim precedes FAIL

E9 #7 already reports D before #8 reports B. We cannot infer that the later FAIL caused the earlier reported correction, or reconstruct actual file state from these messages.

A useful challenge/reconsideration episode is not the same as a proven accuracy improvement caused by verification.

Evidence: E9 #2, E9 #5, E9 #6, E9 #7, E9 #8, E9 #10, E9 #11, E9 #12, E9 #13. A future harness should timestamp artifact versions and actual stop events: the transcript continues after an early PASS despite a stated PASS-based stopping rule.

Turn plausible mechanisms into a controlled, externally scored pilot before training

CURRENT LIMITS

What this cohort cannot establish

A small, convenience-selected set with repeated tasks, two descriptive solo/team pairs, mixed Scout versions, and partial visibility does not identify general failure rates or causal gains.

There are no independent grades, file-write logs, token/cost records, or validated human labels. Public message span is not active runtime. Task-only silence is unobservable behavior, not a diagnosed stall.

Training and broad reported team experiments are not claimed as completed by this analysis.

PROPOSED NEXT PILOT — NOT RUN

Test ownership, not just extra prompting

For coding tasks, compare a solo baseline, a fixed-role team baseline, and the same team with explicit task/artifact claims plus accepted ownership transfer.

Match tasks, model/version, aggregate per-run budget, seeds, and external grader. Keep roles and verifier policy fixed between the two team conditions. Repeat conditions; separate exploratory and held-out tasks.

Start with a feasibility pilot, then set sample size and budget from observed variability; this report does not claim statistical power.

PRIMARY OUTCOME

External task success

Use a task-specific grader and preserve the produced artifacts; do not substitute self-reported PASS.

MECHANISM & COST

Ownership + execution evidence

Log artifact writers/versions, ownership transitions, useful versus redundant checks, tokens, elapsed time, and stop events.

ANALYSIS VALIDITY

Human adjudication first

Two reviewers independently label sampled episodes. Adjudicate disagreements; assess per-class errors and agreement before treating Scout as a research measure.

> **Proposed decision gate:** only pursue a training target after the intervention improves independently scored outcomes under controlled budgets and the detector's judgments survive human review.

Every result links to frozen evidence

The source evidence links are pinned to data commit `fd0be2ee134c2a355de55f3442c167652840bedd`. Each row links the four selected completed runs. The full Scout run ID is printed; all run IDs are also in the generated summary.json.

REFERENCE / SOURCE SLUG	CANONICAL TRACE / SOURCE	SELECTED ANALYSES / SCOUT RUN
E1 aziz-bundle-n4-r2	`trace_464e5df863013053` snapshot / transcript / SharedNet	demo, regex, metrics, Scout `run_05ffc3afe6c94700982201b9e220832c`
E2 aziz-bundle-n4-r1	`trace_e82952d9d64878ba` snapshot / transcript / SharedNet	demo, regex, metrics, Scout `run_fd882547bd5d4af499fd3829defc37d4`
E3 aziz-bundle-n3-r2	`trace_039e550dde384d68` snapshot / transcript / SharedNet	demo, regex, metrics, Scout `run_f6aefa71129e4c90b02b517e71f39152`
E4 aziz-merger-n3	`trace_6faa8fd53c465dab` snapshot / transcript / SharedNet	demo, regex, metrics, Scout `run_211327e58c894fe18eb884e0edbc2bc9`
E5 xiangqian-noop007i2-solo	`trace_ab1efd80c85a6001` snapshot / transcript / SharedNet	demo, regex, metrics, Scout `run_ec9c12ad964d4b2a94bb2e13a75529a5`
E6 xiangqian-noop007i2-pev	`trace_29d100b4dfa5eb0d` snapshot / transcript / SharedNet	demo, regex, metrics, Scout `run_9d2d44edca7440a7914b7cc051bb37ad`
E7 xiangqian-noop-stalled	`trace_34cf0b34ad7096e4` snapshot / transcript / SharedNet	demo, regex, metrics, Scout `run_c4de7935f4c045c3b68fa1a00ef321b5`
E8 xiangqian-gpqa027-solo	`trace_b682600c5b65b12d` snapshot / transcript / SharedNet	demo, regex, metrics, Scout `run_d7b71d27ce23419b84e9179606943b13`
E9 xiangqian-gpqa027-pev	`trace_5c5209c98f666281` snapshot / transcript / SharedNet	demo, regex, metrics, Scout `run_531686d965eb48d79da4f2dca4b433cd`

The archived JSON snapshots and transcripts remain usable if a live share changes. Original SharedNet links are provenance, not our sole evidence. No Xisen traces or `xisen-w/rac` were analyzed in this cohort.

Task prompts change the metrics; complete post coverage is not complete text visibility

Measured p95 message length, in words

TRACE	ALL POSTS	AGENT ONLY	EVIDENCE
E1 Bundle / 4 agents / r2	1,603	19	all posts / agent only
E2 Bundle / 4 agents / r1	1,603	34	all posts / agent only
E3 Bundle / 3 agents / r2	1,603	23	all posts / agent only

Long operator/task prompts dominate the all-post p95 in these three bundle traces. This does not show that agents sent very long messages. Agent-only E7 metrics are undefined, not invented zeros. Solo dominance is expected from the condition, not evidence of a collaboration failure.

> **Metric interpretation:** adjacency and turn switching are not proof of a handoff or information uptake. Metrics describe visible conversation structure, not task success.

Bounded text and history

Per annotation: up to 6 explicitly cited earlier messages at 2,000 characters each, 12 preceding messages at 500 each, and 8,000 target characters. No future messages. Collaboration: up to 80 candidates at 2,000 characters each.

70 candidates across the cohort; 0 traces hit candidate-count truncation. However, 3 source messages exceed the target-text cap and 3 exceed the candidate-text cap. The saved `truncated=false` does not certify unabridged text.

Model-judged primary message types

TYPE	POSTS
result delivery	26
coordination	14
delegation request	10
status update	9
acknowledgement	3
critique disagreement	2
proposal	2
agreement	1
artifact sharing	1
clarification	1
decision	1

All 70 posts, including 11 task/operator posts. One primary type per post; up to two secondary types. Of 15 possible primary labels, only nonzero labels are shown. This is not an agent-only distribution or a validated communication taxonomy measurement.

Scout's categories and confidence remain provisional until human annotation and per-class validation. Citation validity alone is not ground truth.

Source definitions: METRICS.md / MESSAGE_ANNOTATION.md

APPENDIX C / CLAIM-TO-MESSAGE LEDGER

Observed statements are not verified execution

REF	TRACE-GROUNDED OBSERVATION	ORIGINAL MESSAGE EVIDENCE
E1	Competing splits; agents 3 and 4 both report finishing tasks 5-7.	E1 #2, E1 #3, E1 #4, E1 #5, E1 #6, E1 #7
E2	Task-3 overwrite warning, restoration, then another overwrite claim.	E2 #12, E2 #15, E2 #16, E2 #17
E3	An overwrite and restoration of recover.json are reported.	E3 #6, E3 #9, E3 #10
E4	Multiple agents report producing the same output files.	E4 #2, E4 #3, E4 #4, E4 #5, E4 #6, E4 #7
E5	Reports answer 10 and its own PASS; no independent grade.	E5 #2
E6	Selects whole-pack interpretation and answer 11 after asking the user.	E6 #2, E6 #4, E6 #5, E6 #6
E7	Only the task is visible; no observable agent activity.	E7 #1
E8	Reports answer B and its own PASS; no independent grade.	E8 #2
E9	Reported answer D to B to D; conflicting file-state reports.	E9 #2, E9 #6, E9 #7, E9 #8, E9 #11, E9 #12, E9 #13

Reported context, not a grade: teammate notes report 5/7 success for E2, 7/7 for E3, reference answer 10 for arithmetic, and D for GPQA. These notes are preserved in the manifest; they were not independently evaluated here.

Interpretive boundary: the three case studies are illustrative readings of this cohort, not human-adjudicated labels or evidence of a general protocol effect. File writes, execution order, private reasoning, and external grades remain unobserved.

For every Scout category judgment, follow its JSON link in Appendix A to inspect the exact model explanation, confidence, coverage, and `message_numbers`. The saved raw labels and historical failures have not been rewritten to match this narrative.

Original turn numbers above are not Scout prompt display labels such as [M#].

Reproduce the report and explore the saved results

1 / Report & saved AgentTrace results

Read HTML/PDF offline, without a server or model key. Rebuild using Python 3.10+: `python3 reports/2026-10-16-progress/build_report.py`; add `--check` to verify saved HTML/JSON.

For the interactive UI, clone/install using the repository setup, then `make build`. From the repository root, with any previous AgentTrace server stopped:

```
source .venv/bin/activate
export AGENTTRACE_DATA_DIR="$PWD/.agenttrace/teammates"
mkdir -p .agenttrace
mkdir -m 700 "$AGENTTRACE_DATA_DIR" &&
  cp data/teammate-traces/agenttrace.sqlite3 "$AGENTTRACE_DATA_DIR/" &&
  cp -R data/teammate-traces/runs "$AGENTTRACE_DATA_DIR/" &&
  agenttrace
```

The non-recursive `mkdir` refuses to overwrite a previous restoration. Never point the API at the committed snapshot. Open `http://127.0.0.1:3000/` and `data/teammate-traces/batch/links.html`.

2 / Native Scout, second terminal

```
source .venv/bin/activate
DATA_DIR="$PWD/.agenttrace/teammates"
mkdir -p "$DATA_DIR/scout-view/transcripts" \
  "$DATA_DIR/scout-view/scans"
scout view --transcripts "$DATA_DIR/scout-view/transcripts" \
  --scans "$DATA_DIR/scout-view/scans" \
  --host 127.0.0.1 --port 7576
```

Click **Open in Scout** in a saved result: replay uses `mockllm/model`, not provider inference. **Do not click Run Scout** to view existing results; a new analysis can incur charges. Keep services on loopback.

Selected results versus all saved history

ANALYZER	STATUS	RUNS
agenttrace_heuristics_v1	completed	9
example_kickstart	completed	10
regex_v1	completed	13
scout_collaboration_v1	completed	9
scout_collaboration_v1	failed	8

49 historical runs, including 8 failed Scout attempts. Select **36 completed analyses** from `batch/index.json`: 18 substantive offline, 9 Scout, 9 demo. Selected completion: 2026-10-12 to 2026-10-14.

Scout versions are not uniform

ANALYZER VERSION	PROMPT	TRACES
2.0.0	2	1
2.1.0	3	8

All use `openai/gpt-5.4-mini`; E7 is the older run. Original generation-event logs and token accounting were not captured.

Code: `2a118aac8bdbc1eed77bed47fc432146770d65e8`
Data: `fd0be2ee134c2a355de55f3442c167652840bedd`

SHA-256 / manifest.json
`47aff0a8c8a611a1deab037b450d7952000bd88f9a56a55203a6e684be3b32c0`
SHA-256 / frozen SQLite
`eb3ee63bae46f8a1cf7167e23a48a04f617cd03546c034fe6f39f257eb0c4e60`

On macOS verify the frozen files with `(cd data/teammate-traces && shasum -a 256 -c SHA256SUMS)`; Linux: `sha256sum -c SHA256SUMS`. Complete safe restore details are in the dataset guide.